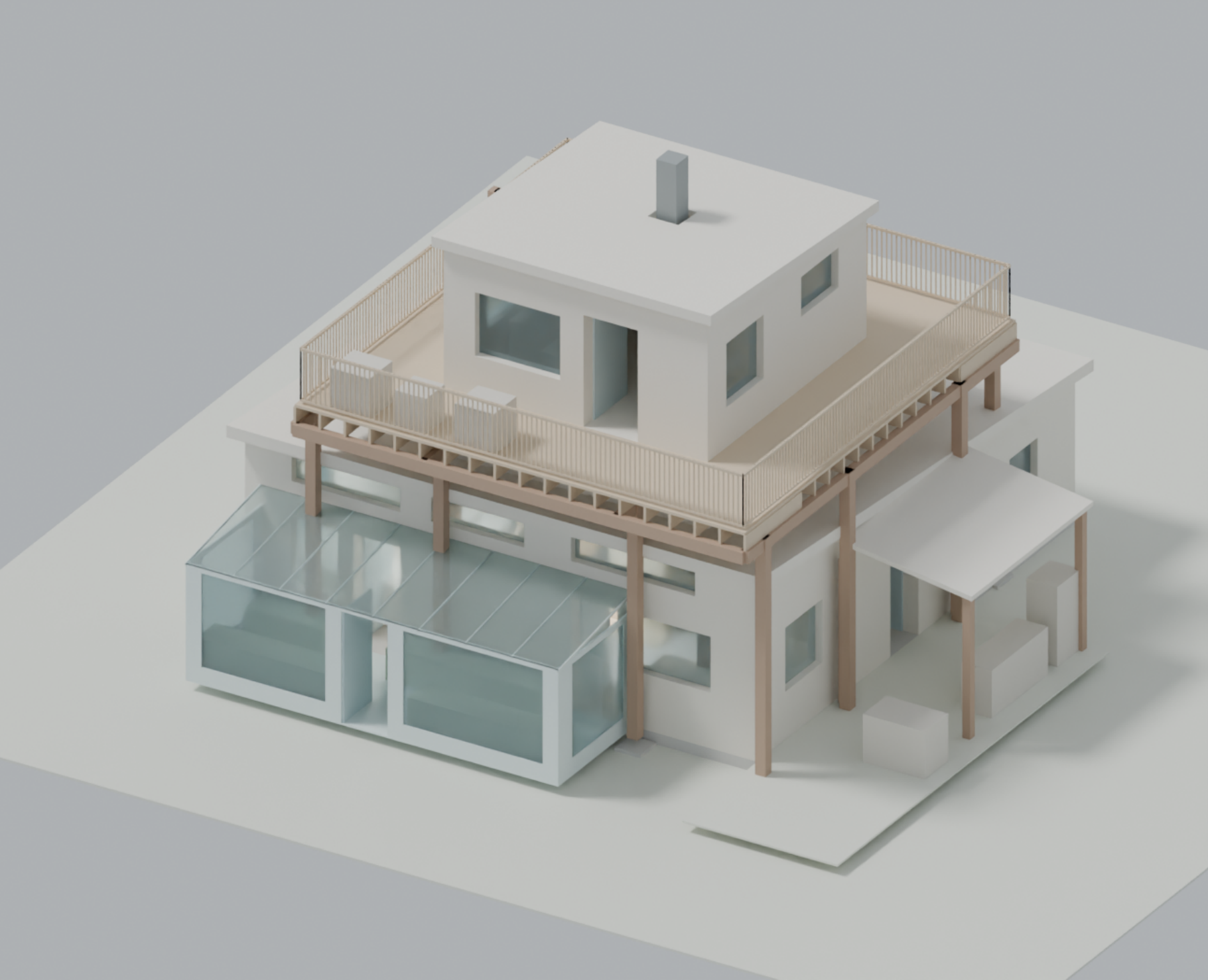


One house from three branches

A first design package for Daniel Irving. Dimensions, bedroom program and foundation remain proposals.



The recommended arrangement

Two bedrooms and all daily services on the main floor. A separate upper work / reading room sits over the hearth and stair. A full deck ring surrounds it.

Part	Proposed size
Main enclosure	33 x 34 ft / 1,122 sq ft
Tower enclosure	17 x 17 ft / 289 sq ft footprint
Upper usable floor	143 sq ft after openings
Deck ring	6 ft nominal / 552 sq ft
Greenhouse	24 x 8 ft / 192 sq ft

The deck is substantial: almost half the main-floor footprint. Its support, roof penetrations, snow exposure and waterproofing drive complexity.

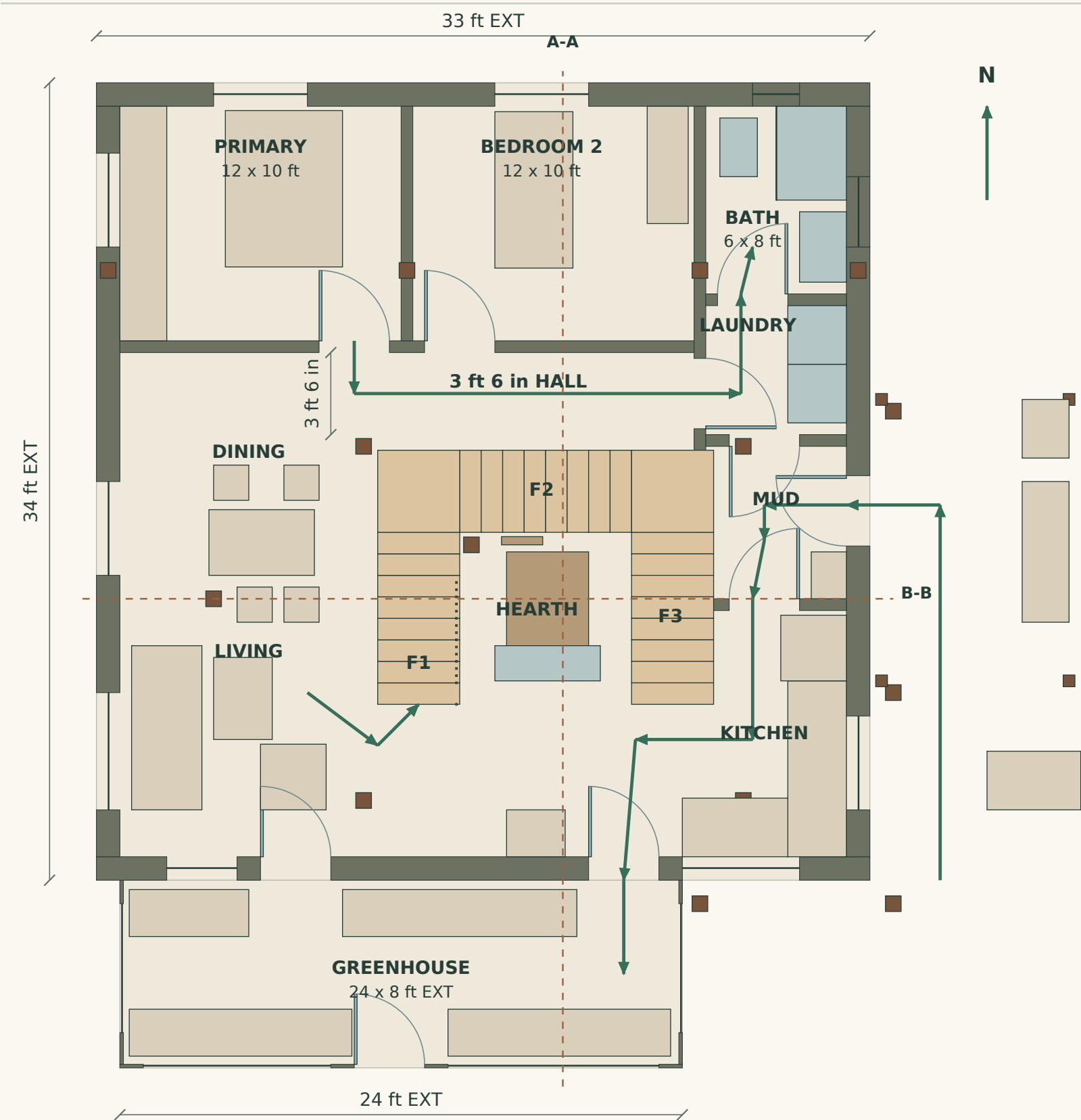
Composition: FL0 v4/v5 | Hearth/stair lessons: v10.9-v10.12 | Full ring: revision 681

This is a new, traceable synthesis. It preserves the south greenhouse, earthen hearth, exposed timber and upper deck destination. It does not adopt the 36 x 36 ft, three-storey shell or the 30.18 ft deck height from revision 681.

HEARTHTREE / H01

A-101 Main floor

North up. Dimensions shown are clear internal dimensions unless marked EXT. Exterior walls 12 in; partitions 6 in.



A daily life that stays downstairs

Arrival enters the mudroom from the sheltered east side. The kitchen reaches the greenhouse through a closable glazed door. Bedrooms have their own hall; neither becomes a passage.

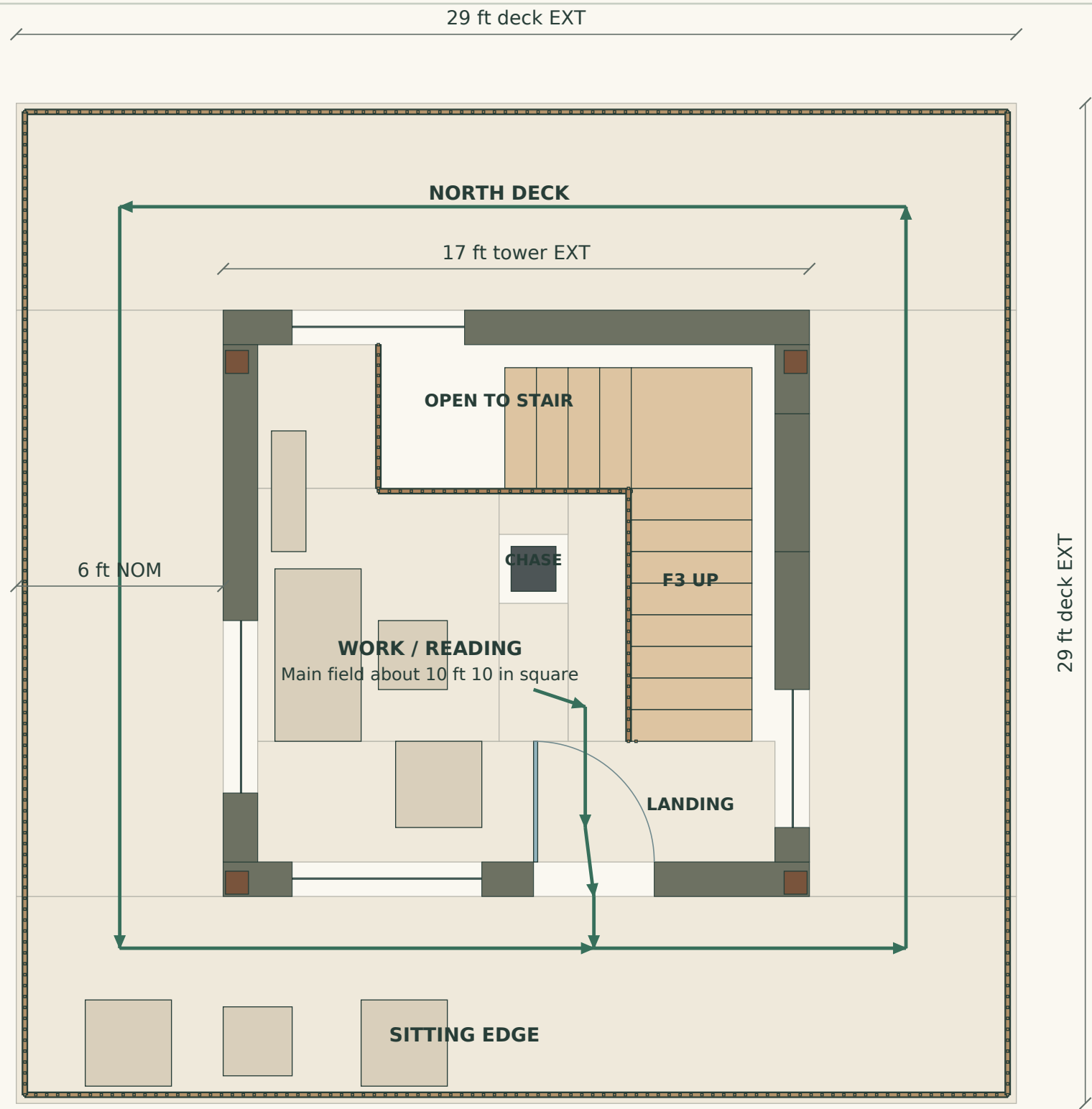
Space	Clear size / fit
Primary bedroom	12 x 10 ft; queen and 2 ft wardrobe
Bedroom 2	12 x 10 ft; single / guest bed shown
Living + dining	10 ft 6 in x 17 ft 6 in open zone
Open kitchen zone	11 x 10 ft 6 in; pantry on south wall
Mud entry	6 x 6 ft 6 in; bench and coat storage
Laundry / services	6 x 5 ft 6 in; shared passage to bath
Bathroom	6 x 8 ft; 3 x 4 ft shower
Bedroom hall	3 ft 6 in nominal; doors 3 ft nominal

Green arrows are sampled circulation routes. Furniture blocks are sized objects in the model. F2 and F3 are projected above the main floor; they do not occupy its whole area.

The bath is reached through a short shared laundry/service passage. Equipment-door operation and turning space need a more detailed pass. The main-floor program is proposed; the latest larger app save contains three bedrooms.

A-102 Tower and continuous ring deck

Floor and deck at +16 ft. This is the upper destination, with a real floor, stair opening, door and guards.



The ring stays usable

Four joined deck fields make one level surface. There are no steps or internal guardrails at the corners. The south tower door opens inward.

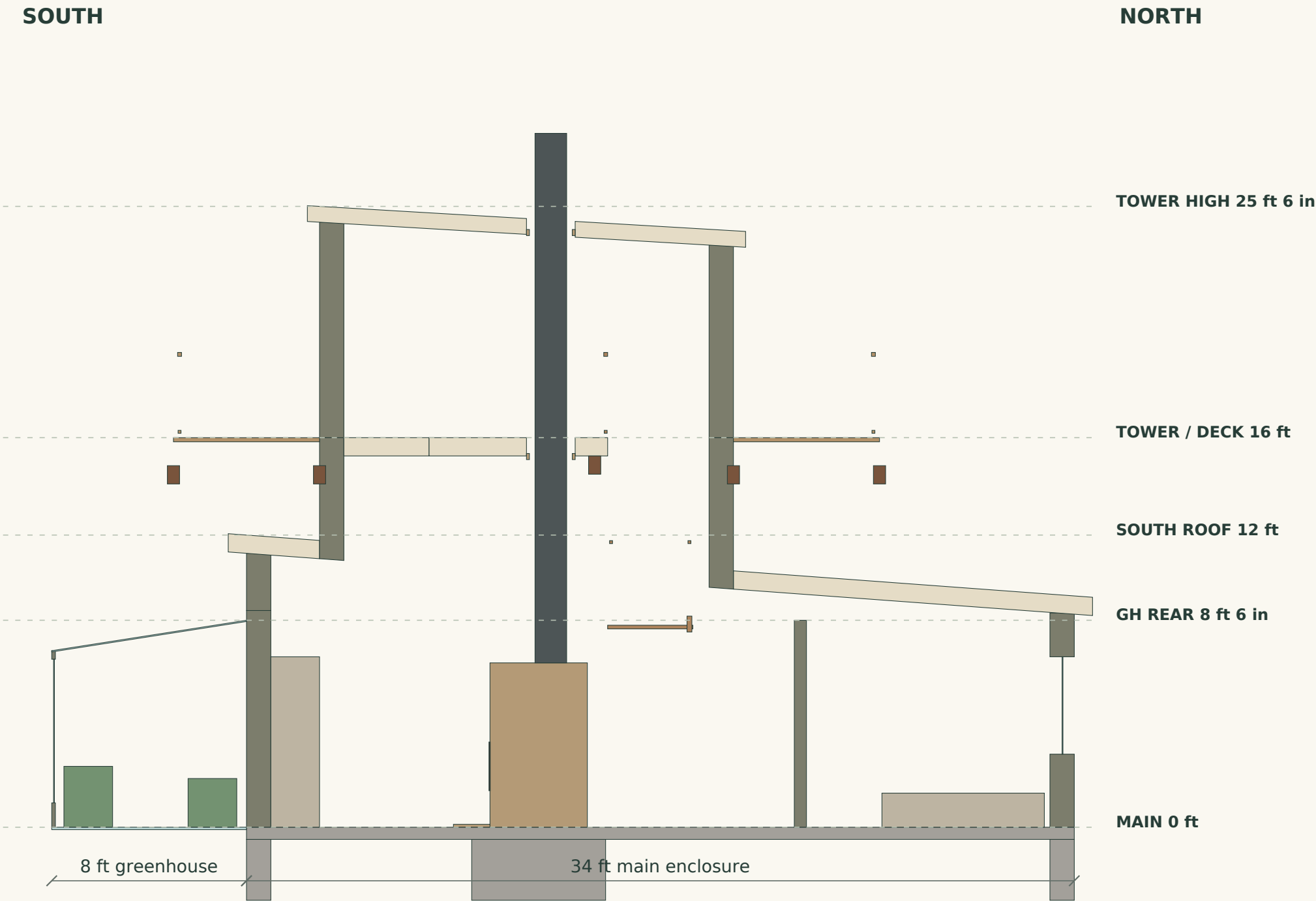
Check	H01 geometry
Deck depth	6 ft nominal; about 5 ft 8 in inside guard
Seating	2 ft 6 in chairs leave 3 ft at inner edge
Upper floor	143 sq ft inside after stair/chimney voids
Door	3 ft 6 in nominal opening
Stair landings	3 ft 6 in square
Deck threshold	Same modeled level; drainage detail pending

The floor opening follows the north and east stair flights. The west return sits above the low first flight, where there is ample modeled clearance.

The 17 ft enclosure is larger than the early 10 x 8 ft tower because a comfortable stair and a useful room take real space. There is no separate full middle storey.

A-301 Section A-A: greenhouse, hearth, tower

True north-south mesh section at X = 19.9 ft. South is left. Stair flight 2 is cut; full stair sequence is on A-302.



Vertical decisions

The deck clears the highest sampled main-roof point below its beams by 2.1 ft. This creates a service space between roof and deck, not another living level.

The chimney is straight and has actual floor and roof openings. The heater, decorative earthen fin, timber bearer and chimney are separate objects.

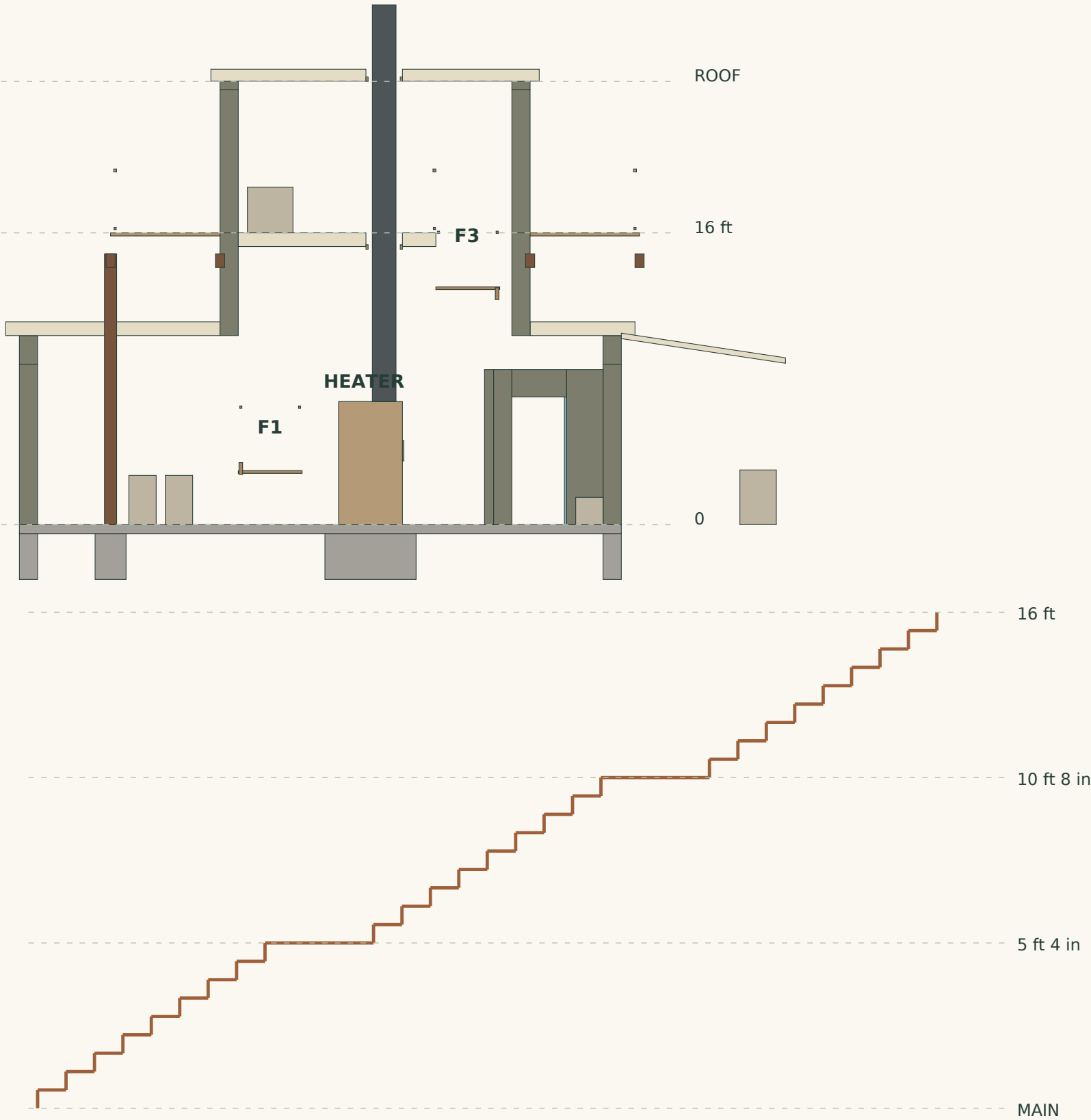
A 4 ft clear zone is reserved in front of the heater. This is a planning allowance, not a verified appliance clearance.

The foundation shown is a support reservation. A basement or walkout could replace it after slope, drainage, groundwater and access are known.

The greenhouse closes off from the house and vents independently. Its roof meets the rear wall at 8 ft 6 in; the house clerestory starts at 9 ft 3 in. Greenhouse shading, vent sizing and moisture control remain design work.

A-302 The hearth and three-flight stair

Section B-B at Y = 12 ft, plus an unfolded profile from the same tread and landing levels.



UNFOLDED WALKING PROFILE / landings are horizontal; no winders

Consistent geometry

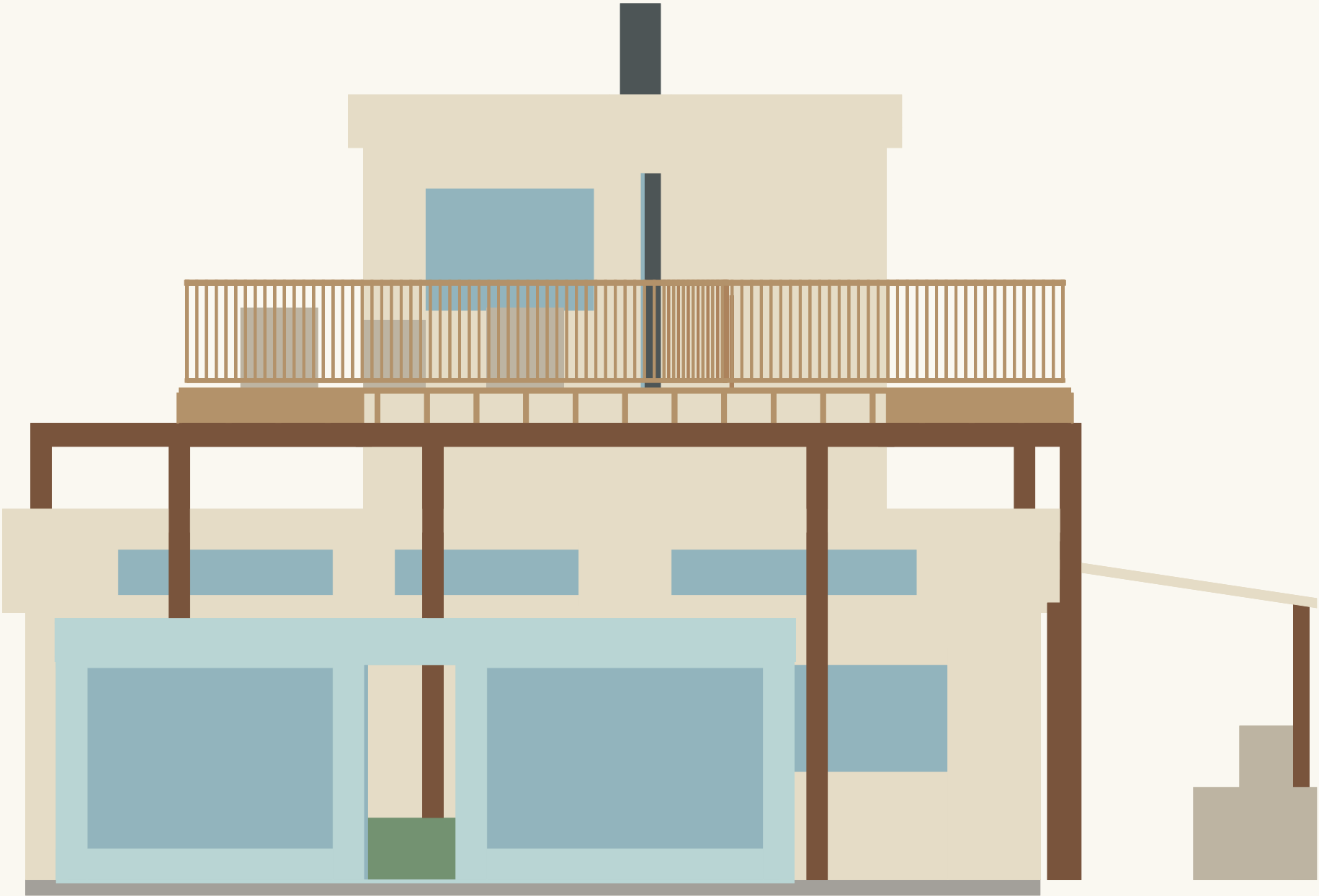
Item	Modeled result
Risers	27 total: 9 + 9 + 9
Each riser	7.11 in
Each going	11 in
Stair width	42 in nominal; 37.2 in between rails
Intermediate landings	42 x 42 in at +5 ft 4 in and +10 ft 8 in
Tread headroom	Lowest sampled: 8.63 ft
Sampling	243 tread/landing samples against actual overhead faces

The tower floor is genuinely cut away above the upper flights. It is not a transparent marker drawn over a solid slab. A south landing connects the last riser to the room and deck door.

The profile proves rise, run and level agreement. Final nosings, guard connections, handrail transitions, landing details and adopted-code review remain open. The sculptural fin is an early massing placeholder.

A-201 South elevation and roof direction

Orthographic projection of the same model. Main house and tower both drain north in the recommended scheme.



Glazed growing space

Upper south glazing

Sheltered arrival / work

Criterion	South-high shed: recommended	East-high shed: comparison
Daylight + greenhouse	Continuous south clerestory above the greenhouse roof.	West end loses much of that upper glazing band; staggered openings needed.
Tower + deck	Full ring works at +16 ft; main roof stays below it.	Ring also clears; changing slope alone does not reduce the stair climb.
Construction + drainage	Two north-draining roof planes; keep runoff away from south growing edge.	Drainage shifts west; greenhouse junction varies across the facade.

Evidence, changes and the next decisions

The accessible source files were inspected; archival claims are distinguished from verified geometry.

Source inspected	Retained	Changed / unresolved
PC v4 README + generation script; Drive v5 complete PDF	South-high shed, southeast kitchen, isolated greenhouse, explicit tower / perch composition.	v4 script raises tower to 14 ft; v5 PDF calls it 10 ft. Older stove substitution is superseded by this brief.
Drive v10.9 / v10.10 / v10.11 notes; v10.12 geometry JSON and actual .blend	Central heater/stair relationship; real flights, landings, voids and separate chimney.	230 mesh objects inspected in Blender. The earlier L-shaped perch survives in v10.12; it is not a full ring. Stair geometry is reconstructed.
PC current-project r681 and GitHub engine at 986836c	Four connected 6 ft deck fields around an upper work room.	Resolved all four deck tops to 30.18 ft. H01 reduces this to 16 ft; the 36 x 36 ft, three-storey shell is not adopted.

Established preferences

South growing space; central masonry hearth and tree-like expression; timber and natural finishes; useful upper tower with surrounding deck; everyday main-level living.

Recommendations in this package

Two main-level bedrooms; one full bathroom; three equal stair flights; a simple 12 in envelope allowance; low main shed with a separate tower; a full ring at +16 ft; independent east work shelter. Straw bale can be substituted only after rechecking interior dimensions and openings.

Your choices / technical review

Confirm whether two bedrooms meet the household requirement, or whether a third must be designed now. Review the tower and deck size before detailed furnishings. Foundation, basement/walkout, heater configuration, structure, fire clearances, low-slope roofing, drainage and mechanical systems need site/product/professional inputs.

What was checked

Common-source mesh drawings; 243 stair/headroom samples; 738 route-center samples using a 24 in body envelope; deck level and area; 4 ft hearth-front zone; greenhouse/clerestory heights. Route obstructions in the final sample: 0. This is a bounded geometry review, not exhaustive circulation, structural, fire or accessibility validation.

Files and source paths

Hearthtree-H01.blend + parameters.json + geometry.json + build_geometry.py + build_blender.py + check_geometry.py + make_package.py. The source/decision record contains complete PC paths and Drive/GitHub links. Original app state and archived design files were not overwritten.

Workflow reference: developers.openai.com/blog/architectural-visualization-with-astra. Its editable geometry / preview / correction approach informed this package; its architecture was not used as a style template. Detailed materials, the final camera tour and interactive walkthrough follow layout selection.